

RUTTVIK KHOLLAM

Senior Software Engineer | React.js | Next.js | TypeScript | Performance & Accessibility Specialist
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Senior Software Engineer with 4+ years of experience building scalable, high-performance web applications using React.js, Next.js, and TypeScript. Specialized in frontend architecture, accessibility (WCAG 2.1), SEO optimization, and performance engineering, consistently achieving Lighthouse scores above 95. Experienced in designing reusable component systems, integrating third-party platforms, optimizing Core Web Vitals, and delivering enterprise-grade SaaS applications. Strong focus on clean code practices, modular architecture, and scalable frontend development.

Work experience

Senior Software Engineer, Invimatic Technologies

May 2022 - Present

- Led frontend development for multiple enterprise-grade SaaS applications using React.js, Next.js, and TypeScript, improving development efficiency through reusable component architecture and modular design systems.
 - Lifted Lighthouse performance score from 35 to 96+ by implementing route-based code splitting, dynamic imports, and srcSet image optimization — reducing First Contentful Paint (FCP) by ~58% on slow 3G connections and lowering landing page bounce rate by ~20%.
 - Reduced initial JavaScript bundle size from ~1.8 MB to ~620 KB through dynamic imports and tree-shaking — improving Time-to-Interactive (TTI) by ~3.2 seconds on slow 3G/4G networks and optimizing Core Web Vitals for low-bandwidth users.
 - Raised Lighthouse accessibility score from 71 to 95+ across 4+ enterprise product surfaces by implementing semantic HTML, ARIA landmarks, and keyboard navigation — helping meet WCAG 2.1 accessibility compliance requirements for client-facing platforms.
 - Integrated third-party platforms including Stripe, EaseBuzz, analytics providers, and authentication services to streamline payments, event tracking, and user workflows.
 - Enforced code quality standards using ESLint, Prettier, and Husky pre-commit hooks, reducing merge conflicts and improving maintainability across projects.
 - Worked on secure authentication flows, token handling, and frontend security best practices including HTTPS, CSP, and XSS protection.
 - Built real-time communication proof-of-concepts using WebSockets and Server-Sent Events (SSE) for live updates and chat functionality.
 - Designed polling systems for long-running backend processes, improving user experience during asynchronous AI-driven content generation workflows.
 - Ran weekly frontend knowledge-sharing sessions on React performance, accessibility, and TypeScript patterns; attended by interns and junior developers across teams.
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Projects

E-KYC Platform for the Hospitality Industry:

- Designed and developed a digital KYC verification platform using Aadhaar image verification and AI-powered face-matching APIs.
- Integrated EaseBuzz payment gateway for secure hotel booking and transaction processing.
- Implemented accessibility-first UI architecture with screen reader compatibility and contrast optimization.
- Built scalable frontend modules using React.js and TypeScript.

Ad Content Automation Platform (Shopify Merchants)

- Developed a multichannel marketing automation platform supporting TikTok, Facebook, Instagram, Mailchimp, and Pinterest campaign generation.
- Integrated PostHog analytics for real-time funnel tracking and user behavior analysis.
- Improved performance for media-heavy catalogs using image optimization, lazy loading, and responsive asset delivery.
- Built scalable reusable UI components for campaign workflows and merchant dashboards.

HRMS System:

- Developed and maintained a full-featured HRMS platform supporting employee onboarding, client management, payroll, project management, timesheets, and leave tracking.
- Integrated Stripe payout workflows for payroll processing.
- Used TanStack Query for efficient API caching and state synchronization.
- Improved maintainability through reusable component structures and strict code quality enforcement.

White-Label E-commerce Platform:

- Built a customizable e-commerce platform with advanced coupon systems, sales analytics, and retailer-specific branding.
- Improved SEO and Core Web Vitals through SSR optimization, lazy loading, and route-level code splitting.
- Enhanced accessibility using semantic HTML, keyboard navigation, and ARIA patterns.
- Optimized frontend architecture for scalability and maintainability.

Skills

Frontend Technologies: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS

State & Data Management: Redux Toolkit, TanStack Query (React Query), Context API, GraphQL

Performance & Optimization: Core Web Vitals, Lighthouse Optimization, SSR / CSR, Dynamic Imports, Route-based Code Splitting, Lazy Loading, Image Optimization, Bundle Size Optimization, SEO Optimization, Webpack Analyzer

Accessibility & UI Engineering: WCAG 2.1, Accessibility, Semantic HTML, ARIA, Keyboard Navigation, Screen Reader Testing, Responsive Design, Component Architecture, Design Systems

Backend & Integrations: Node.js, NestJS, REST APIs, MongoDB, IntegrationBus, WebSockets, Server-Sent Events (SSE), Stripe, EaseBuzz, Authentication & Session Handling

Security: JWT Authentication, HTTPS, CSP, XSS/CSRF Protection

Tools & Platforms: Git, GitHub, ESLint, Prettier, Husky, PostHog, Google Analytics

Development Practices: Agile/Scrum, Modular Architecture, Clean Code, Performance Optimization, Third-Party Integrations, Client Requirements Analysis

Testing & Quality: Jest, React Testing Library, Vitest, Playwright/Cypress, Component Testing, Mock Service Worker (MSW)

Education: BE (Mechanical) — Dr. D.Y. Patil School of Engineering, Pune | 2020

Profiles

Github: <https://github.com/ruttvik1021>

Portfolio: <https://ruttvikkhollam.in>

LinkedIn: <https://in.linkedin.com/in/ruttvik1021>